

Dylan Kawalec

AI Solutions Architect (Confidential Systems)

Forward Deployed Engineering | Applied Research | Protocol Integrations

Building the trust layer for private intelligent systems

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Professional Summary

AI Solutions Architect for confidential systems. I build production private AI runtimes, TEE-backed deployment paths, and protocol integrations, then use research artifacts to validate and formalize the underlying architecture. Recent work includes Phala dstack deployments for 24+ partner teams, Rust SGX tooling for Osmosis on Azure, and zero-trust verifier prototypes in Python/C++. Most roles were contract or founding engagements across early-stage infrastructure teams.

Core Competencies

Technologies: Python, Rust, C/C++, Go, TypeScript, Solidity, PyTorch, FastAPI, Docker, GitHub Actions, Postgres, Redis.

AI Systems: local-first inference (Ollama, llama.cpp, vLLM; Llama/Mixtral/Qwen), agent orchestration, verifier pipelines.

Confidential Compute: Intel SGX/TDX, Phala Cloud/dstack, Fortanix, RA-TLS, remote attestation, zero-trust execution.

Infrastructure & Protocols: AWS, Azure SGX VMs, Cloudflare, ZK proofs, threshold signing, blockchain systems, TLA+/PlusCal.

Production Systems & Research Artifacts

- **Production:** Mermate, Synth, POEMS, dstack integrations (Phala), [Warp Engine](#), and Akave confidential gateway work for private AI deployments.
- **Research:** SGX decentralized infrastructure ([defe](#)), zero-trust verification systems (Cosmic/ENI6MA), and [DI](#) for ERC-8004 agent identity.

Experience

Head of AI, Decentralized Confidential Compute *Jokuh (Independent R&D)* Dec 2025 – Present

- Designed and deployed private AI systems for agent orchestration and local-first inference across TEE and cloud environments.
- Built internal tooling to generate architectures, APIs, and deployment pipelines for multi-agent systems.
- Led development of confidential AI workflows enabling secure execution, verification, and data isolation.

Head of Developer Relations & Solutions Architect (Protocol Integrations) *Phala Network* May – Dec 2025

- Led confidential AI rollout on Intel TDX, enabling secure deployments across 24+ partner teams.
- Designed integrations with EigenLayer, Rift.trade, Akave, and Redpill.ai for TEE-based execution.
- Built Docker deployment workflows and hands-on integration support that grew dstack visibility 34→400+ GitHub stars.

Principal Research Engineer *Cosmic Inc. | ENI6MA Research* Summer 2024 – Jan 2026

- Designed zero-trust verification systems for confidential compute and deterministic execution.
- Built Python/C++ verifiers and proof pipelines for validating execution inside secure environments.
- Delivered provisional patent architecture supporting a \$168K friends-and-family raise.

Head of Developer Relations (Protocol Integrations) *GOAT Network (BTC L2)* Jul 2024 – Jan 2025

- Delivered Founders Club MVP and onboarded 100+ founders; authored zkMIPS/EVM docs, ran BTC L2 workshops, and supported ZKM-led Bitcoin L2 integration research.

Research Engineer *Osmosis Foundation* May – Jul 2024

- Built SGX-based infrastructure tooling in Rust for Fortanix deployment on Azure.
- Developed [defe](#), a CLI for TEE lifecycle management, enclave TLS, key lifecycle, and threshold signing.

CTO & Principal Engineer *Invo Technologies* Apr 2023 – Feb 2024

- Built the Invo MVP on AWS; completed patent architecture, led github.com/Invo-Technologies/poems, and won Aleo Best Tech 2023 (\$50K grant).

Earlier: Led integrations and ecosystem development across BNB Chain Labs, Findora/Discreet Labs, Harmony, Tesla, and Zygo, working across cryptography, blockchain infrastructure, developer systems, manufacturing, and precision optics.

Recognition, Credentials & Publications

Awards: ETHGlobal SF 2024 (Ledger 1st, Polygon 1st), Aleo Best Tech 2023, VeChain×EasyA PowerUp (2nd), ETHDenver 2024 Hacker Deathmatch.

Credentials: RareSkills Research Fellow; ZKU Developer; B.S. Computer Science, WGU (2019–2021); A.A.S. Computer Programming, MiraCosta (2015–2019).

Selected links: phalanetwork.mintlify.app/docs | docs.goat.network | eni6ma.gitbook.io/eni6ma | github.com/DylanCkawalec/cosmic-hack | blackbox-podcast.com